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Challenging Ultra Slim Wellbore Reach Potential Unlocked: Tailored, Chemically Bonded Centralizers Enables to Reach Deeper Zones – A Success Story from Western Offshore India

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Abstract

Cementing is one of the most important steps when drilling and completing oil or gas wells, and well cementing technology is the application of many scientific and engineering disciplines. The objective of the primary cement operation is to provide zonal isolation and effective cement placement. To achieve this result, one must engage in the following practices including good drilling practices and mud properties, casing movement while cementing, centralization of the casing, optimal borehole pipe clearance, and using spacers and mud flushes.

Cementing within slimhole and ultra-slimhole wells causes additional challenges—from running casing/liner to cementing operations, and solid-type centralizers molded on casing surface help overcome these challenges. Objectives include:

- Centralizing within ultra-slim hole environments, such as 5-in. production liners within 6-in. open holes
- Achieving long-term production using zonal isolation
- Achieving well integrity by delivering optimized standoff for a successful cementing operation
- Reducing the potential risk of failure and the time for running and landing casing to total depth
- Achieving uninterrupted fluid flow along the annulus to reduce equivalent circulation densities (ECDs)
- Cost-effective alternative to conventional centralizer subs

The process includes:

- Comparing conventional and solid-type centralizers molded on casing surface after simulation with well data
- Optimizing the position of these molded centralizers on casings using standoff calculator and three-dimensional (3D) displacement efficiency simulations

- Establishing a make-up procedure and QA/QC process

Solid-type centralizers molded on the casing surface have been used for 20+ wells within an ultra-slimhole 5-in. liner (6-in. openhole). Conclusions include:

- Drastic reduction in torque and drag while running-in-hole of casing
- Dramatic decrease for ECDs during circulation and cement operation
- Contributes to well integrity by facilitating even distribution of cement for full coverage—satisfactory cement bond log/variable density log records

Introduction

The Mumbai High field is located 160 km W-NW offshore from Mumbai in Western Coastal shelf of India at a water depth of ~75m and is the biggest field in India, with an aerial extent of 1200 sq km (V. Agrawal et al. 2008) ¹.

Sediments deposited in Mumbai High field (figure 1) over a period ranging from the Paleocene to recent times are dominated by sea level changes. Both clastic and carbonated sediments deposited are hydrocarbon rich with good reservoir quality developed in most places. The field is a carbonate reservoir of the Miocene age, primarily composed of multi-layered limestone with interbedded shale. The structure is a doubly plunging anticline with faulted limbs, which acts as a structural trap (Debasish Dasgupta et al. 2008) ².



Figure 1—Location of Mumbai High

Due to continuous production over several decades, the main reservoirs have undergone significant depletion. As a result, operators have been venturing into deeper horizons and untapped zones to enhance hydrocarbon recovery and sustain production levels

Well Architecture

Wells in Mumbai High are typically directional or high angle (figure 2) to maximize reservoir contact. The casing design is tailored to handle:

- Conductor Casing: 20" or 30" set at shallow depths (~100–150 m) to support marine risers.
- Surface Casing: 13-3/8" set at ~500–800 m to isolate shallow formations and provide blowout preventer (BOP) support.

- Intermediate Casing: 9-5/8" or 10-3/4" set across problematic formations (e.g., depleted zones or lost circulation zones).
- Production Casing/Liner: 7" or 5-1/2", depending on the completion strategy.

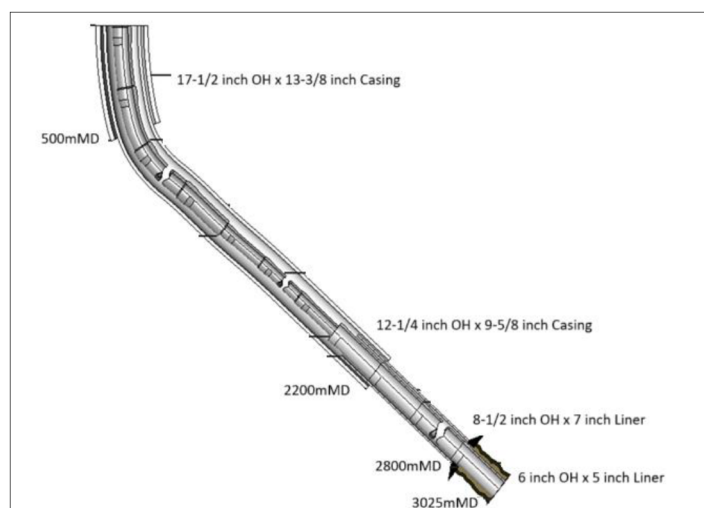


Figure 2—Well architecture

Drilling Challenges

- Reservoir Depletion: Long-term production has led to pressure depletion, increasing the risk of differential sticking and wellbore instability
- Severe Lost Circulation: Especially in fractured carbonates, leading to non-productive time (NPT) and increased mud costs.
- Complex Geology: Heterogeneous reservoir with multiple flow units and fault blocks complicates well placement and reservoir management.
- Water Cut Management: High water production in mature wells necessitates zonal isolation and selective perforation strategies

Cementing in slim holes with low annular clearance presents several technical and operational challenges that can significantly impact zonal isolation, well integrity, and overall success of the cementing job. Few of the key challenges are listed below.

Narrow Annular clearance

- Reduced Mud Displacement Efficiency: The small gap between casing and borehole wall makes it difficult to effectively displace drilling mud with spacer and cement, increasing the risk of mud channeling.
- Higher Friction Pressure: Pumping fluids through a narrow annulus requires higher pressures, which can exceed formation fracture gradients, especially in weak formations.
- Limited Flow Paths: Turbulent flow is harder to achieve, reducing the effectiveness of mud removal and leading to poor cement bonding

Centralization Difficulties

- Narrow Pressure Window: The Equivalent Circulating Density (ECD) must be carefully managed to avoid fracturing the formation or inducing losses.

- Rheology Optimization: Slurry must be engineered for low friction and good displacement efficiency while maintaining adequate thickening time and compressive strength.
- Risk of fluid Migration: In low-clearance annuli, the risk of unwanted fluid migration increases due to poor mud removal and low hydrostatic pressure margins.

Operational Risks

- Loss Circulation: High ECDs can lead to losses, especially in depleted or fractured zones.
- Stuck Pipe or Casing: Tight clearances increase the risk of casing getting stuck during running or cementing operations.

Remediation Challenges

- Remedial Cementing Complexity: Squeeze jobs are more difficult to execute effectively due to limited access and pressure control.

The technical and operational complexities associated with cementing in slim holes with low annular clearance underscore the need for innovative approaches to ensure well integrity. The restricted annular space not only hampers effective mud displacement and cement placement but also limits the deployment of conventional centralization tools. These limitations can lead to eccentric casing, poor zonal isolation, and increased risk of operational failures. Therefore, there is a critical need to adopt alternative centralization strategies.

Methodology

This study adopts a structured approach to evaluate the effectiveness of rigid-resin centralizers in slimhole wells with complex geometries and high dogleg severity. The methodology is designed to assess both the mechanical and hydraulic performance of these centralizers compared to conventional solutions. The key steps are as follows:

Selection and Deployment of Rigid-Resin Centralizers

Rigid-resin centralizers were selected for their ability to maintain structural integrity under high dogleg severity and tight annular clearances. These centralizers were deployed in field wells known for challenging trajectories to minimize nonproductive time (NPT) during casing operations.

Circulation Pressure Evaluation

Circulation pressures were monitored and analyzed to assess the impact of centralization on fluid displacement efficiency. This involved comparing pressure profiles with and without the use of rigid-resin centralizers to determine improvements in hydraulic performance.

Torque and Drag Analysis

Torque and drag simulations were conducted using casing running models. Field data was collected to compare the mechanical resistance encountered during casing deployment with and without rigid-resin centralizers. This helped quantify the reduction in mechanical friction and the ease of casing placement.

Centralization and Standoff Comparison

The centralization performance of rigid-resin centralizers was benchmarked against traditional centralizers. Standoff ratios and displacement efficiency were evaluated using simulation software and validated with field measurements to determine improvements in cement placement quality.

High-Strength Composite Centralizer

In ultra slim hole drilling environments, where annular clearances are minimal and wellbore geometries are often complex, achieving effective casing centralization is a critical challenge. Traditional steel or bow-spring centralizers may not provide the necessary flexibility or durability under high dogleg severity and tight restrictions. To address these limitations, centralizers engineered from advanced resin-composite materials are robust alternative. These centralizers offer a unique combination of rigidity, wear resistance, and low-profile design, making them particularly well-suited for slimhole applications. Their ability to maintain casing standoff, reduce drag, and withstand mechanical stresses during casing deployment contribute significantly to improved cement placement and zonal isolation.

The installation (Fig. 3) of resin-composite centralizers involves a precise and controlled process to ensure optimal bonding and performance in ultra slimhole applications. These centralizers are directly molded onto the casing surface, eliminating the need for mechanical attachment and allowing for a low-profile design tailored to the well's specific geometry. The process begins with surface preparation, where the designated area on the casing is sandblasted to create a clean, dry, and contaminant-free surface. This step is critical to achieving the required anchor profile, with a minimum surface roughness of 80 microns to ensure strong adhesion of the resin material.



Figure 3—Resin-composite centralizer bonding process

Following surface preparation, custom-designed molds—manufactured based on the well's centralization requirements—are clamped onto the casing. The mold edges are carefully sealed to prevent leakage. Using a precision injection system, the resin-composite material is then pumped into the mold cavity. Once the mold is completely filled, it is sealed and left undisturbed to allow the resin to cure and harden under controlled conditions.

After curing, the mold is removed, and the centralizer undergoes a series of quality control checks. These include measuring the blade height to ensure it meets design specifications and verifying the outer diameter (OD) using a calibrated ring gauge. Additionally, each blade is tested for hardness, with a target value of 80 Shore D. If the required hardness is not achieved, the centralizer is subjected to additional curing

time to reach the desired mechanical properties. This meticulous process ensures that each resin-composite centralizer delivers consistent performance in terms of standoff, durability, and resistance to deformation during casing operations in high dogleg and narrow annular environments

Circulation Pressure Evaluation

Effective fluid displacement during cementing is critical for achieving zonal isolation and long-term well integrity. One of the key indicators of displacement efficiency is the circulation pressure profile observed during the pumping of spacer and cement slurries. In this context, the use of resin-based composite centralizers has shown measurable improvements in hydraulic performance, particularly in wells with complex geometries or tight annular clearances.

One of the defining advantages of resin-based composite centralizers lies in their strategic geometric profile and equidistant placement along the casing string. As illustrated in the design schematic (Fig. 3), these centralizers are installed at precisely spaced intervals, with an offset angle of 120° around the pipe's circumference. This configuration ensures uniform standoff and balanced annular flow paths throughout the wellbore.

Unlike solid centralizers, which often obstruct significant portions of the annular space due to their bulk and continuous contact with the casing, resin-based centralizers are engineered with streamlined blades or ribs that maintain minimal surface contact while maximizing fluid bypass area. The 120° offset angle further enhances this effect by distributing the centralizers evenly around the pipe, preventing flow channeling and dead zones.

Hydraulic Benefits:

- Increased annular flow area: The open-profile design allows for more unobstructed fluid movement around the casing.
- Improved mud displacement: Uniform standoff and flow symmetry reduce the risk of mud channeling and residual pockets.
- Lower circulation pressures: With less resistance to flow, the system experiences smoother pressure profiles and reduced ECD.
- Enhanced cement sheath quality: Better displacement leads to more complete mud removal and improved bonding between casing and formation.

Resin-based composite centralizers offer a significant improvement in managing Equivalent Circulating Density (ECD), particularly in formations with narrow fracture gradients. Unlike solid centralizers, which tend to restrict flow due to their bulkier design and reduced flow-by area, composite centralizers are engineered to minimize flow resistance. This design enhancement leads to smoother pressure profiles during circulation and displacement operations.

Figure 4 illustrates this advantage clearly. The ECD measured in the fracture zone using resin composite centralizers consistently remains lower than that observed with conventional solid centralizers. The difference in ECD reaches up to 0.03 sg, which is critical in tight-margin wells where even minor pressure increases can lead to fracture initiation or lost circulation. This reduction in ECD is attributed to improved flow dynamics around the casing due to the streamlined geometry of composite centralizers.

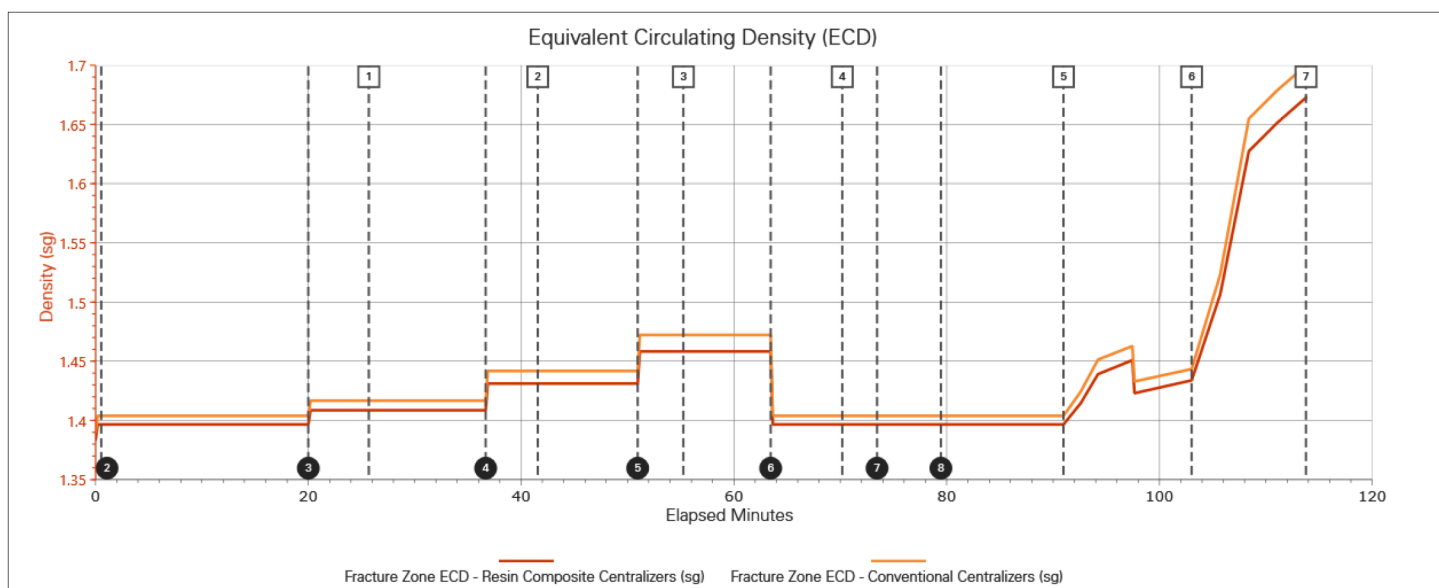


Figure 4—Comparison of ECD's with resin composite centralizers vs conventional centralizer

This design philosophy not only improves displacement efficiency but also contributes to better cementing outcomes, particularly in deviated or horizontal wells where flow dynamics are more complex.

Torque and Drag Analysis

In the evolving landscape of well construction, particularly in extended-reach and high-angle wells, managing torque and drag is a critical operational challenge. Traditional solid centralizers—typically made from steel or aluminum—offer high mechanical strength but often contribute to elevated frictional resistance, increased casing wear, and higher surface torque. In contrast, resin composite centralizers have demonstrated significant advantages in reducing torque and drag, improving operational efficiency, and enhancing cementing outcomes.

Low Coefficient of Friction

Resin composite centralizers, especially those made from resin based composite materials, exhibit exceptionally low coefficients of friction as low as 0.089 (Urdaneta Javier et al. 2008)³. This reduction in friction translates directly into:

- Lower surface torque during casing rotation.
- Reduced drag forces during casing running.
- Easier casing deployment in tortuous or deviated wellbores.

This is particularly beneficial in extended-reach drilling (ERD) and horizontal wells, where frictional forces can otherwise prevent casing from reaching total depth.

Enhanced Wear Resistance and Durability

Resin composite centralizers can withstand side loads and outperform any metal-based centralizers. Their wear resistance ensures consistent performance over long runs, reducing the need for reaming back and forth.

Corrosion Resistance and Formation Compatibility

Being non-metallic, resin composite centralizers are immune to corrosion and non-damaging to CRA (corrosion-resistant alloy) casing. They also reduce the risk of galvanic corrosion and are formation-friendly, especially in soft or reactive formations where steel centralizers may cause damage.

Improved Torque Consistency

Unlike metallic centralizers, which can exhibit high static friction, resin composite centralizers maintain a minimal difference between static and dynamic friction. This results in smoother casing rotation and more predictable torque profiles, which is crucial for real-time torque-and-drag modeling. During the casing running, the torque reduction of approximate 10% in torque values are observed compared to solid type centralizer. Figure 5 illustrates the torque values comparing with solid centralizers and without centralizers.

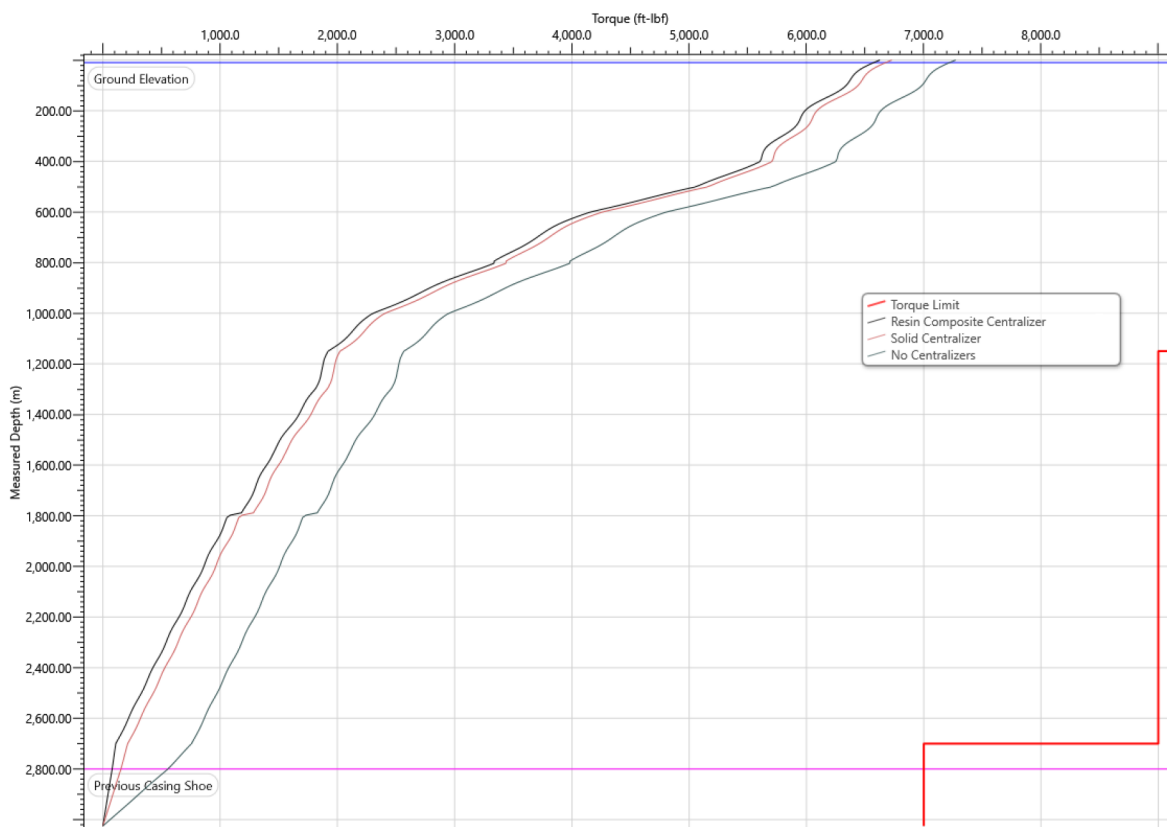


Figure 5—Torque reduction comparisons

Centralization and Standoff Comparison

Centralization is a critical factor in achieving effective cement placement and long-term zonal isolation. The standoff ratio, defined as the percentage of the annular space maintained between the casing and the borehole wall, directly influences the uniformity of fluid flow and the displacement of drilling mud. In this study, the performance of rigid-resin composite centralizers was rigorously evaluated against traditional metallic and bow-spring centralizers through simulation modeling.

In this study, simulation modeling was used to compare the standoff profiles of resin composite centralizers against conventional metallic centralizers. The results, illustrated in Figures 6(a) and 6(b), clearly demonstrate the superior performance of resin-based centralizers:

- The resin composite centralizers maintain a higher and more consistent standoff across the measured depth (MD) range, with values generally staying above 70%, and often approaching 100% at centralizer locations.
- In contrast, conventional centralizers show greater variability in standoff, with values frequently dropping below 70%, especially at midpoints between centralizers.
- The standoff at midpoints—a critical indicator of centralization effectiveness—remains significantly higher with resin composite centralizers, reducing the risk of mud channeling and poor cement bonding.

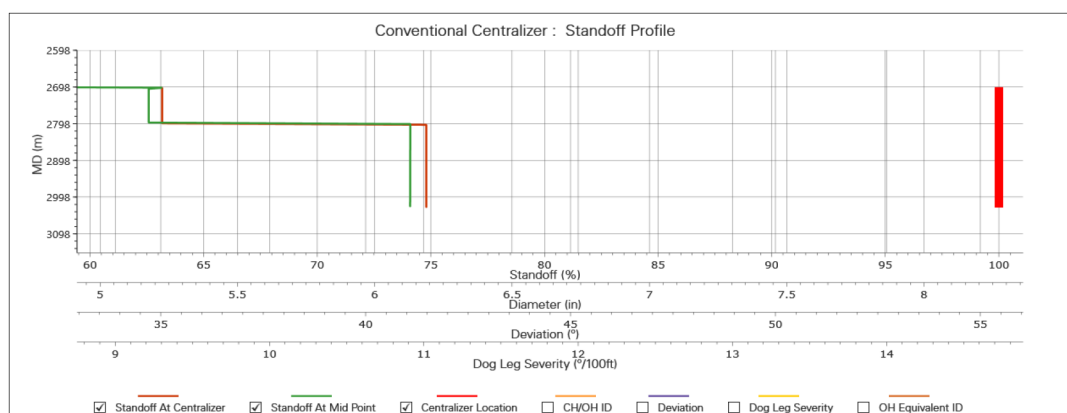


Figure 6(a)—Standoff profile with conventional centralizer

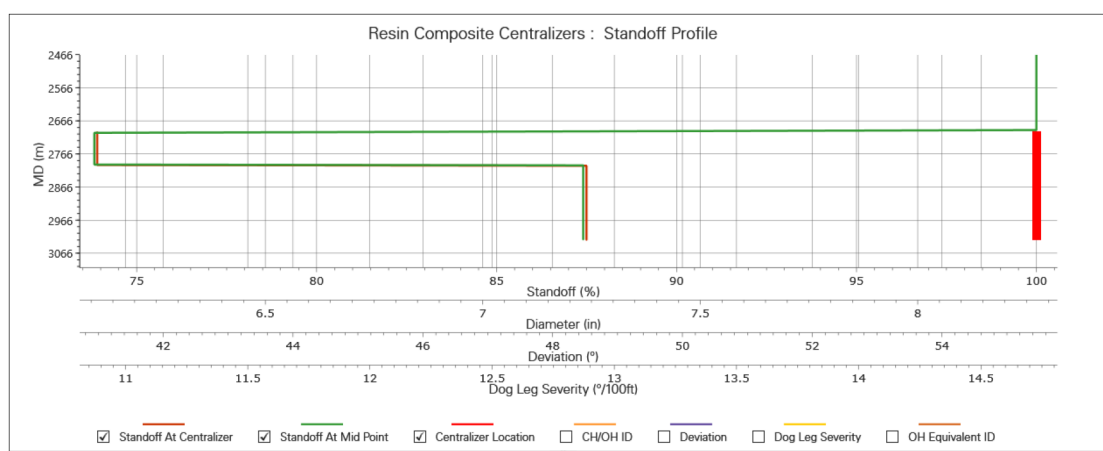


Figure 6(b)—Standoff profile with resin composite centralizers

This improved performance is attributed to:

- The rigid structure of resin composites, which resists deformation under downhole loads.
- Optimized blade geometry, which ensures better wall contact and casing support.
- Lightweight yet durable material properties, allowing for better deployment in high-deviation or high-DLS (Dog Leg Severity) sections

By maintaining optimal standoff, resin-based centralizers enhance cement sheath integrity, reduce the likelihood of micro-annuli formation, and contribute to the long-term reliability of zonal isolation.

Displacement Efficiency

Displacement efficiency is a critical parameter in cementing operations, directly influencing the quality of mud removal and the integrity of the cement sheath. Achieving high displacement efficiency ensures that drilling fluids are effectively displaced by the spacer and cement, minimizing the risk of channeling, poor bonding, and long-term zonal isolation failure.

To optimize this process, the use of advanced 3D computer simulation tools is highly recommended. These simulators allow engineers to model various centralizer configurations and evaluate their impact on standoff, fluid dynamics, and cement placement. One of the primary functions of centralizers is to reduce casing eccentricity, which in turn minimizes the disparity in annular flow area and frictional forces between the narrow and wide sides of the annulus. This is particularly important in highly deviated and horizontal wells, where gravitational forces tend to pull the casing toward the low side of the borehole, increasing the risk of poor displacement and cement channeling.

A major challenge in such environments is understanding the interaction of fluids downhole. In eccentric annuli, fluid velocities differ significantly between the narrow and wide sides, leading to uneven displacement fronts. This can compromise the top of cement (TOC), result in annular pressure buildup, and necessitate costly remedial operations such as squeeze cementing—or in worst cases, lead to well abandonment.

Effective monitoring and modeling are essential to track how fluid interfaces evolve over time within the wellbore. In this study, a three-dimensional displacement simulation was conducted to evaluate the performance of rigid-resin composite centralizers [Fig. 7](#). The results demonstrated that these centralizers improved the displacement efficiency to 67%, particularly in complex wellbore geometries where achieving cement displacement efficiency is challenging. The improved centralization led to more uniform fluid fronts, better mud removal, and a more continuous cement sheath—key factors in achieving long-term well integrity.

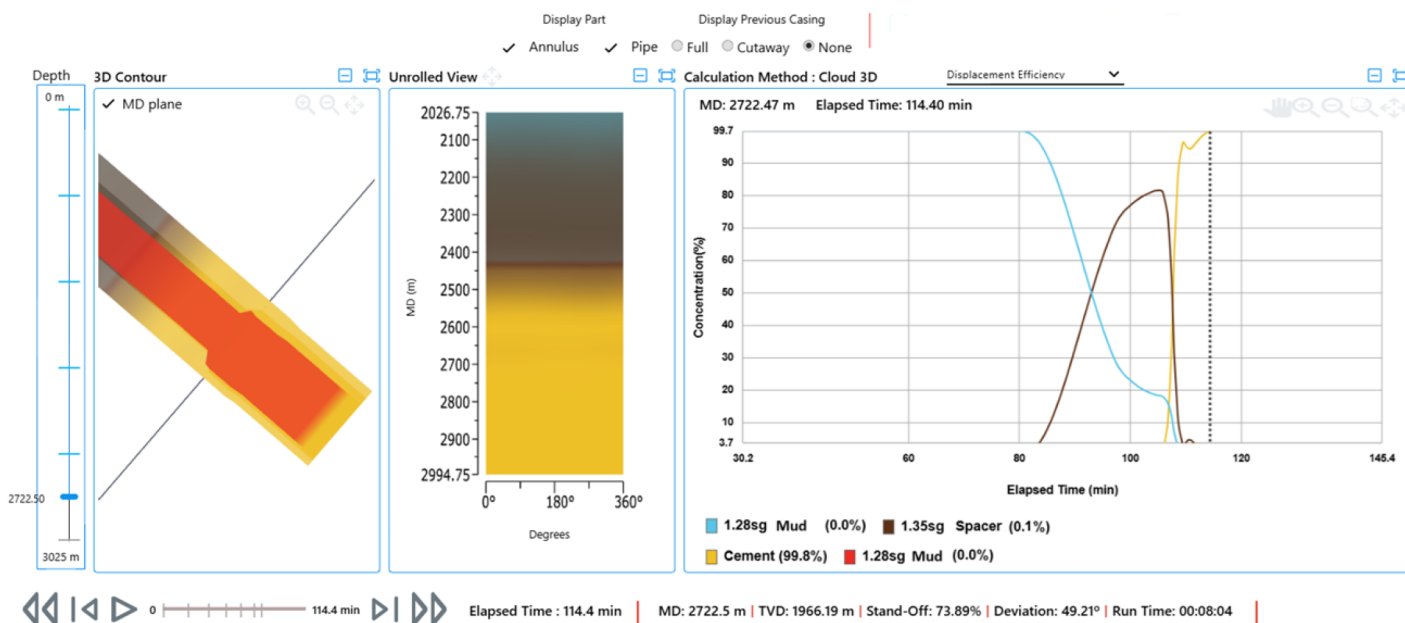


Figure 7—Displacement efficiency of cement in annulus

Conclusion

The deployment of high-strength resin-based composite centralizers represents a significant advancement in wellbore centralization technology, particularly in ultra-slimhole drilling environments where traditional centralizers often fall short. These composite centralizers offer a unique combination of mechanical

robustness, low-profile geometry, and enhanced hydraulic performance, making them ideally suited for complex wellbore conditions characterized by tight annular clearances, high dogleg severity, and extended-reach trajectories.

One of the most compelling advantages of resin-composite centralizers is their customizable installation process, which involves direct molding onto the casing surface. This eliminates the need for mechanical attachments and allows for precise tailoring to the well's geometry. The meticulous surface preparation, mold design, resin injection, and curing process ensure that each centralizer meets stringent quality standards in terms of blade height, outer diameter, and material hardness. These attributes contribute to consistent standoff performance, which is critical for achieving effective cement placement and long-term zonal isolation.

From a hydraulic perspective, resin-based centralizers significantly improve circulation pressure profiles during cementing operations. Their streamlined blade design and strategic placement—typically at 120° offsets around the casing—facilitate balanced annular flow paths, reduce flow resistance, and minimize dead zones. This geometry enhances mud displacement efficiency, leading to better cement bonding and reduced risk of channeling. The reduction in Equivalent Circulating Density (ECD)—up to 0.03 sg compared to conventional solid centralizers—is particularly valuable in formations with narrow fracture gradients, where even minor pressure increases can trigger lost circulation or formation breakdown.

The study's use of three-dimensional displacement simulation further validated the superior performance of resin composite centralizers. The simulations revealed that these centralizers not only maintain higher and more consistent standoff ratios—often exceeding 70% and approaching 100% at centralizer locations—but also deliver improved fluid interface control. This is especially important in eccentric annuli, where uneven fluid velocities can compromise the top of cement (TOC) and lead to annular pressure buildup. By promoting uniform displacement fronts, resin centralizers help mitigate these risks and reduce the likelihood of remedial squeeze operations or well abandonment.

In addition to hydraulic and centralization benefits, resin composite centralizers offer notable improvements in torque and drag management. Their low coefficient of friction—as low as 0.0891—reduces casing running forces, enhances casing reach in extended-reach wells, and minimizes wear on both the casing and wellbore. Unlike metallic centralizers, which often exhibit high static friction and inconsistent torque profiles, resin-based designs maintain a minimal difference between static and dynamic friction. This results in smoother casing rotation, more predictable torque behavior, and improved compatibility with real-time torque-and-drag modeling.

Furthermore, the durability and wear resistance of resin composite centralizers make them a reliable choice for long runs in challenging environments. Their ability to withstand high side loads and resist deformation ensures consistent performance without the need for frequent reaming or repositioning.

In summary, the integration of high-strength resin composite centralizers into well construction workflows offers a multi-dimensional improvement across centralization, hydraulics, mechanical efficiency, and cementing outcomes. Their adaptability to slimhole geometries, superior standoff performance, reduced ECD, and enhanced torque-and-drag characteristics position them as a transformative solution for modern drilling challenges. As well designs continue to evolve toward more complex and constrained geometries, the role of advanced centralization technologies like resin composite centralizers will become increasingly indispensable in ensuring well integrity, operational efficiency, and long-term production success.

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